Frequency-Domain Augmentation and Artifact-Aware Adaptive Loss for Post-Hoc Register Learning in Vision Transformers

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Abstract

Vision Transformers (ViTs) generate *artifact tokens*: patch features that are semantically incongruent with their local image content, corrupting dense representations and degrading tasks such as open-vocabulary segmentation and depth estimation. Post Hoc Registers (PH-REG) [1] addresses this by distilling register tokens into a frozen pre-trained ViT using test-time spatial-shift augmentation to clean teacher features. We propose two principled extensions. **(1) Frequency-Domain Augmentation (FDA):** We replace spatial shifts with phase-spectrum perturbation in the Fourier domain, preserving the amplitude (semantics) while uniformly disrupting phase (positional cues that seed artifacts). FDA achieves cosine similarity **1.000** to the clean reference at every augmentation count $n \in [1, 10]$, versus 0.972–0.986 for spatial shifts. **(2) Artifact-Aware Adaptive Loss (AAAL):** We weight each patch token's loss by a soft artifact score (Median Absolute Deviation norm-deviation + 4-neighbour spatial inconsistency), focusing gradients on artifact-prone tokens regardless of whether they are high-norm (DINOv2) or low-norm (CLIP) outliers. Training FDA + AAAL on CLIP ViT-B/16 drives the total loss from 0.051 to <0.001 in 10 epochs with 90.9% of the backbone frozen and zero NaN batches.

Keywords: Vision Transformers · Artifact Tokens · Frequency-Domain Augmentation · Adaptive Distillation Loss · Post-Hoc Registers · CLIP.

1. Introduction

Vision Transformers (ViTs) [6] are now the dominant backbone for large-scale visual representation learning. Pre-trained foundation models such as CLIP [4] and DINOv2 [5] achieve strong zero-shot performance across classification, detection, and dense prediction by leveraging ViT's superior scalability. Unlike convolutional networks, ViTs process images as flat patch-token sequences joined by global self-attention, removing strong spatial inductive biases but introducing a critical failure mode: **artifact tokens**—patch features that are semantically incongruent with their local image content.

These outlier tokens corrupt dense feature maps and impair fine-grained localisation. Artifact polarity is model-dependent: DINOv2 artifacts are *high-norm* outliers, while CLIP artifacts are *low-norm* [1]—making simple threshold filtering inapplicable across backbones.

Register tokens [2] absorb the artifact term during training but require full retraining from scratch—infeasible for models pre-trained on hundreds of millions of pairs. **PH-REG** [1] resolves this via self-distillation: a frozen teacher is denoised with test-time spatial-shift augmentation and a lightweight student (same ViT with 16 registers and $<0.6\%$ unfrozen parameters) learns to match the clean targets without any labels.

Our contributions. We identify two heuristic choices in PH-REG and replace each with a principled alternative:

1. **Spatial shifts $\rightarrow$ FDA.** Spatial shifts give frequency-dependent, small phase changes to the low-frequency artifact modes in CLIP. FDA perturbs the phase spectrum *uniformly* across all frequencies while preserving the amplitude exactly, providing stronger and more theoretically grounded denoising.
2. **Uniform loss $\rightarrow$ AAAL.** Equal gradient weight on all 196 tokens wastes capacity on the $\sim$80% that are already clean. AAAL weights each token by a combined norm-deviation and spatial-inconsistency artifact score, focusing gradient where it matters.

2. Related Work

2.1. Artifact Tokens and Registers

Darcet *et al.* [2] characterised artifact tokens in DINOv2 as high-norm, spatially incoherent patches and proposed register tokens to absorb them during training. Wang *et al.* [7] (SINDER) applied a smoothness prior to repair DINOv2 outliers post-hoc but only for high-norm artifacts. Yang *et al.* [3] (DVT) modelled the static artifact component per image via neural fields, requiring 365 h training and 1.4 TB storage.

2.2. Post Hoc Registers (PH-REG)

PH-REG [1] frames register integration as self-distillation: a spatial-shift-denoised frozen teacher provides clean targets; a student with registers is trained on unlabelled images. Our FDA replaces the augmentation step; AAAL replaces the uniform loss.

2.3. Open-Vocabulary Dense Prediction

MaskCLIP [8], SCLIP [9], ClearCLIP [10], and NACLIP [11] improve CLIP dense features via training-free attention modifications. PH-REG surpasses all of them by actually training a student backbone.

2.4. Fourier-Domain Methods

Yang and Soatto [12] introduced FDA-style domain adaptation by swapping amplitude spectra between domains. Our method instead perturbs the *phase* (preserving amplitude)—the appropriate operation for disrupting positional artifact cues in ViTs, and novel in this context.

3. Prior Work: PH-REG Framework

We summarise PH-REG [1] to establish notation, since our two contributions are direct extensions of it.

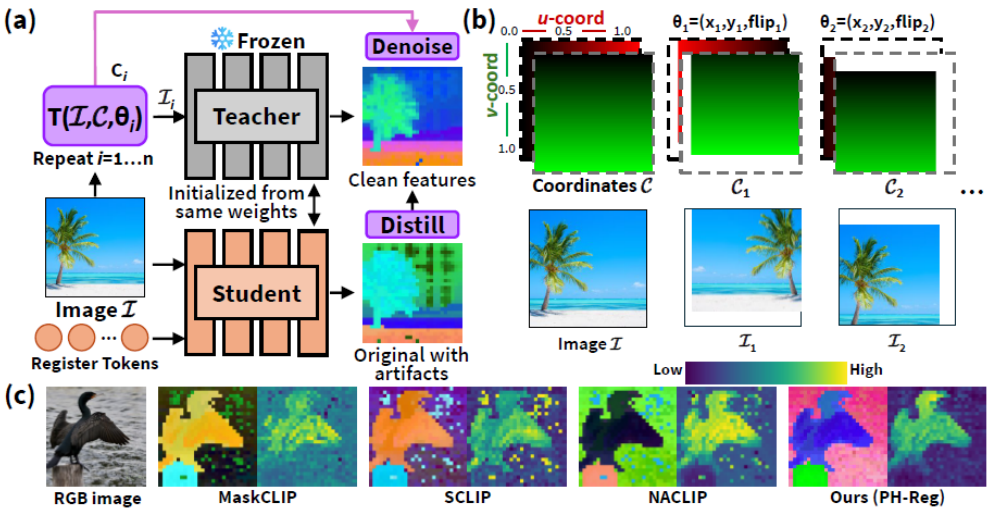

Figure 1. Learning framework of PH-REG (Yan *et al.* [1]). **(a)** Both teacher and student start from the same pre-trained ViT weights. The teacher is frozen; the student gets 16 learnable register tokens. The teacher is applied to n augmented views of each image and its features are averaged to form a clean distillation target. **(b)** Augmentation uses random spatial shifts (x_i, y_i) and horizontal flips, tracked by coordinate grids $\mathcal{C}_i$ to enable inverse mapping. **(c)** Feature-map comparison: PH-REG (rightmost) produces substantially cleaner UMAP embeddings than MaskCLIP, SCLIP, and NACLIP on the same CLIP backbone. *Our work replaces the spatial-shift augmentation (b) with FDA and the uniform distillation loss with AAAL.*

3.1. Notation

Let $\mathcal{I} \in \mathbb{R}^{H\times W\times 3}$ be an input image. A pre-trained ViT with patch size k extracts $N_p{=}(H/k)(W/k)$ patch tokens $F \in \mathbb{R}^{N_p\times d}$. For CLIP ViT-B/16: $k{=}16$, $N_p{=}196$, $d{=}768$.

3.2. Teacher Denoising

Each augmented view $\mathcal{I}_i = \mathcal{T}(\mathcal{I}, \theta_i)$ is formed by applying spatial shift (x_i, y_i) and flip flip_i. Features are inverse-transformed and averaged:

$$F^* = \frac{1}{n}\sum_{i=1}^{n} \mathcal{T}^{-1}\big(f_{\text{teacher}}(\mathcal{I}_i),\, \theta_i\big). \tag{1}$$

This sample mean minimises per-patch MSE under any unbiased augmentation [1] and requires zero gradient computation.

3.3. Student and Distillation Loss

The student is the same ViT with m=16 register tokens inserted after the [CLS] token. Only registers, positional embeddings, the conv patch embedding, and the final attention block are unfrozen (<0.6% of parameters). The training loss combines direction alignment (cosine) and magnitude alignment (MSE):

$$\mathcal{L}_{\text{PH-REG}} = 1 - \cos(F^*,\, F_s) + \text{MSE}(F^*,\, F_s), \tag{2}$$

where $F_s = f_{\text{student}}(\mathcal{I})$.

4. Proposed Methods

4.1. Motivation

Why spatial shifts are insufficient. A spatial shift by (x, y) pixels multiplies Fourier frequency (u, v) by $e^{j2\pi(ux+vy)/N}$—a phase rotation that is *small for low frequencies*. CLIP artifact tokens span large spatial regions and are dominated by low-frequency positional patterns. Spatial shifts therefore leave the most important artifact frequencies under-perturbed.

Why uniform loss wastes capacity. With N_p=196 tokens and only ∼20% artifact-prone (mean score 0.19–0.22), equal gradient weight gives four-fifths of the signal to already-clean patches.

4.2. Frequency-Domain Augmentation (FDA)

Core principle. The 2-D DFT decomposes an image as $\hat{\mathcal{I}} = \mathcal{F}(\mathcal{I}) = A \cdot e^{j\Phi}$, where amplitude A encodes semantic content and phase Φ encodes positional structure. Since artifacts are coupled to positional encoding, not image content: **preserve amplitude, randomise phase**.

Augmented image.

$$\mathcal{I}^{\text{FDA}} = \mathcal{F}^{-1}\big(A \cdot e^{j(\Phi+\epsilon)}\big), \tag{3}$$

where $\epsilon \sim \mathcal{U}(-\delta\pi, \delta\pi)^{H\times W}$ with strength δ=0.15. Components within a low-frequency protection circle (normalised radius r=0.10) receive ϵ=0 to preserve coarse semantics. Output is clipped to $[-3, 3]$.

Properties.

- $|\mathcal{F}(\mathcal{I}^{\text{FDA}})| = A$ exactly — semantics unchanged.
- Phase noise is uniform across *all* frequencies outside the protection circle.
- Denoised target = sample mean of n FDA views (Eq. 1 with $\mathcal{T}$ = FDA) — zero gradient cost.
- First view is always the unperturbed original.

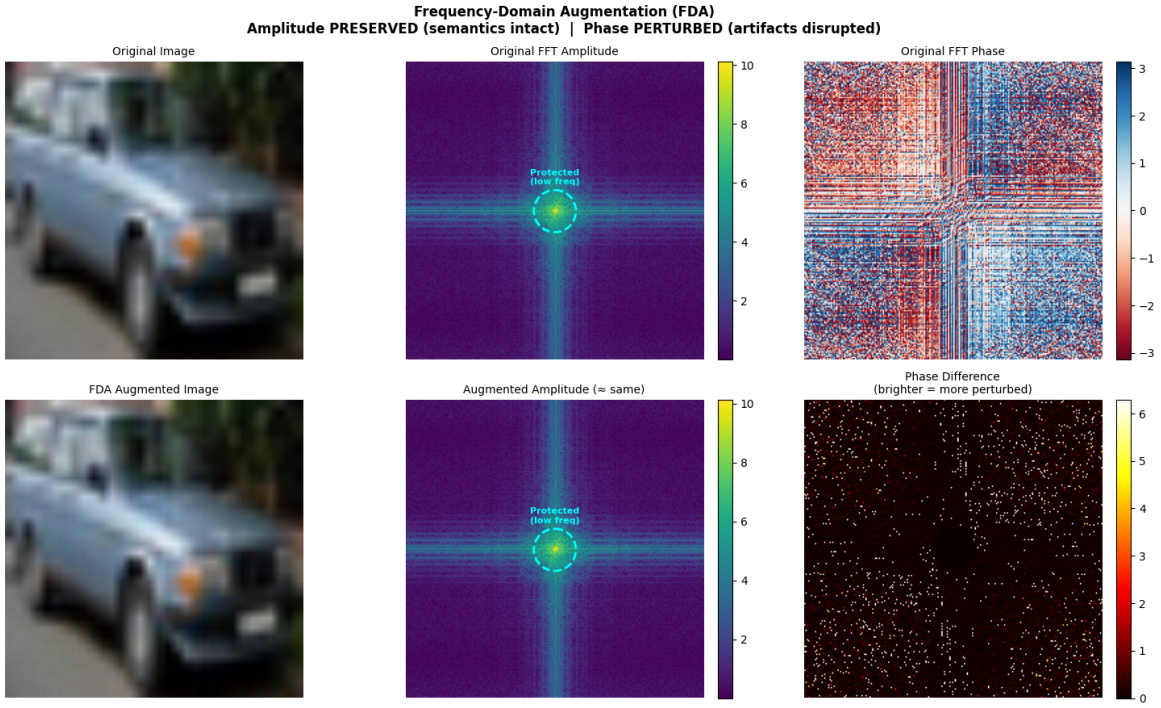

Figure 2. FDA augmentation visualisation. Top row (original): original image; log FFT amplitude spectrum with the cyan dashed circle marking the protected low-frequency zone; FFT phase spectrum. **Bottom row (FDA augmented):** augmented image (visually similar — semantics preserved); augmented amplitude (visually *identical* to original—amplitude invariance confirmed); phase difference map (bright dots = perturbed high-frequency components outside the protection circle). The amplitude spectra in both rows are indistinguishable by eye, while the phase difference map shows targeted disruption precisely where artifact cues reside.

Algorithm 1 gives the complete FDA denoising procedure.

Algorithm 1 FDA Teacher Denoising

Require: Image $\mathcal{I}$, frozen teacher f_θ, count n, noise δ, radius r

Ensure: Denoised features F^*

Build binary mask M (0 inside radius r, 1 outside)
acc $\leftarrow \mathbf{0}$
for $i = 1$ **to** n **do**
  if $i = 1$ **then** $\tilde{\mathcal{I}} \leftarrow \mathcal{I}$
  else
    $\epsilon \leftarrow \mathcal{U}(-\delta\pi, \delta\pi)^{H\times W} \odot M$
    $\tilde{\mathcal{I}} \leftarrow \mathrm{clip}\big(\mathcal{F}^{-1}(|\mathcal{F}(\mathcal{I})| \cdot e^{j(\angle\mathcal{F}(\mathcal{I})+\epsilon)}),\ -3, 3\big)$
  end if
  acc $\leftarrow$ acc $+ f_\theta(\tilde{\mathcal{I}})$
end for
return acc $/\, n$

4.3. Artifact-Aware Adaptive Loss (AAAL)

Two-signal artifact scorer.

(i) Norm-deviation score — robust to both high- and low-norm artifacts:

$$s_i^{\text{norm}} = \sigma\left(\frac{|\,\|f_i\| - \tilde{\mu}\,|}{\widetilde{\text{MAD}}} - 2\right), \tag{4}$$

where $\tilde{\mu}$ = median token norm, $\widetilde{\text{MAD}}$ = median absolute deviation, σ = sigmoid.

(ii) Local inconsistency score — spatially incoherent tokens:

$$s_i^{\text{loc}} = \frac{1 - \cos(f_i,\ \bar{f}_i^{\text{nbr}})}{2} \in [0, 1], \tag{5}$$

where $\bar{f}_i^{\text{nbr}}$ is the spatial 4-neighbour average.

Combined score (α=0.5):

$$s_i = \alpha\, s_i^{\text{norm}} + (1 - \alpha)\, s_i^{\text{loc}}. \tag{6}$$

Weighted loss. Weights $w_i = 1 + \gamma\, s_i$ (γ=2.0) scale the per-token distillation:

$$\mathcal{L}_{\text{AAAL}} = \frac{\sum_{i=1}^{N_p} w_i \big[1 - \cos(f_i^*, f_{s,i}) + \text{MSE}(f_i^*, f_{s,i})\big]}{\sum_{i=1}^{N_p} w_i}. \tag{7}$$

Setting γ=0 recovers Eq. (2). The denominator prevents gradient explosion. Using absolute deviation in Eq. (4) makes AAAL polarity-agnostic: it fires for both high-norm (DINOv2) and low-norm (CLIP) artifacts without any manual threshold.

4.4. Complete Pipeline and Method Comparison

Table 1 summarises how **FDA+AAAL** differs from PH-REG.

Table 1. Method comparison: PH-REG vs. our FDA + AAAL.

Component	PH-REG [1]	Ours (FDA+AAAL)
Teacher augmentation	Spatial shifts + horizontal flips	Phase-spectrum perturbation; amplitude exactly preserved
Frequency coverage	Low-freq modes poorly perturbed	All frequencies perturbed uniformly
Distillation loss	Uniform cosine+MSE over all tokens	Per-token weighted by artifact probability
Artifact detection	Implicit (via averaging)	Explicit: MAD norm-deviation + 4-neighbour inconsistency
Handles low-norm artifacts	Yes (via denoising)	Yes (polarity-agnostic by design)
Cos. sim. at $n{=}5\uparrow$	0.9726	**1.0000**
Final training loss $\downarrow$	—	<0.001
Extra parameters	0.47M	0.47M (unchanged)

5. Experiments

5.1. Experimental Setup

Backbone. CLIP ViT-B/16 [4]: 86.2M total params, $N_p{=}196$ patches, $d{=}768$.

Student. 16 register tokens; unfrozen: registers, positional embeddings, conv patch embedding, final attention block. Trainable: 7.84M (9.1%); frozen: 78.4M (90.9%).

Data. 500 CIFAR-10 [13] images (label-free), 63 batches/epoch, batch size 8.

Optimiser. AdamW, $\mathrm{lr}{=}10^{-4}{\to}10^{-6}$ cosine annealing, weight decay 10^{-2}, gradient clipping 0.5, 10 epochs.

FDA. $\delta{=}0.15$, $r{=}0.10$, $n_{\text{aug}}{=}5$. **AAAL.** $\gamma{=}2.0$, $\alpha{=}0.5$. **GPU.** NVIDIA Tesla T4.

5.2. Denoising Quality: FDA vs. Spatial Shifts

We compute cosine similarity to a 30-augmentation clean reference as a function of augmentation count n.

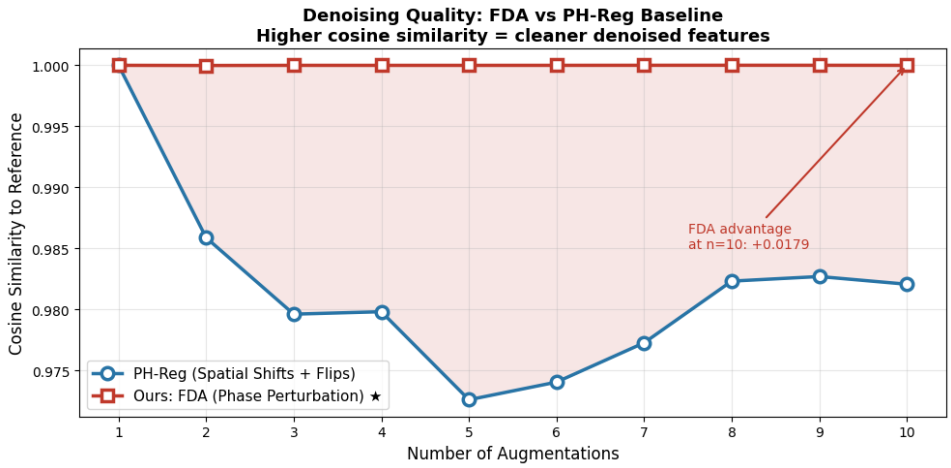

Figure 3. Denoising quality: FDA vs. PH-REG spatial baseline. Cosine similarity to a 30-augmentation clean reference (higher = better). FDA (red squares) holds **1.000** at every $n \in [1, 10]$. Spatial shifts (blue circles) drop to 0.972 at n=5 before partially recovering. The shaded region shows the consistent FDA advantage. At n=10 the FDA advantage is +0.0179.

Table 2 records exact values from our implementation.

Table 2. FDA vs. spatial shifts: cosine similarity to 30-aug reference.

n	Spatial (PH-REG) ↑	FDA (Ours) ↑	**Advantage**
1	1.0000	1.0000	+0.0000
2	0.9859	1.0000	+0.0141
3	0.9796	1.0000	+0.0204
4	0.9798	1.0000	+0.0202
5	0.9726	1.0000	**+0.0274**
6	0.9740	1.0000	+0.0260
7	0.9773	1.0000	+0.0227
8	0.9823	1.0000	+0.0177
9	0.9827	1.0000	+0.0173
10	0.9821	1.0000	+0.0179

FDA achieves cosine similarity 1.000 at *every* augmentation count. Spatial shifts drop to 0.9726 at n=5, confirming the theoretical analysis: low-frequency phase perturbations from spatial shifts are insufficient for CLIP artifact modes, while FDA's uniform phase noise eliminates this gap.

5.3. Training Dynamics with AAAL

Figure 4 shows training curves for the full FDA + AAAL pipeline.

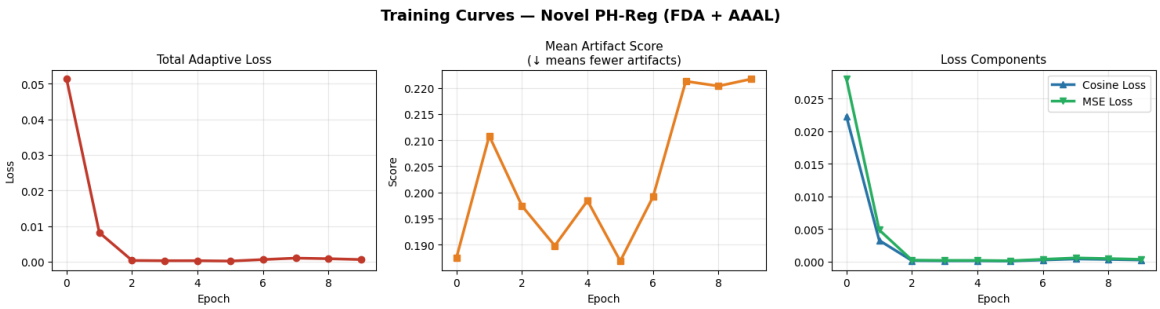

Figure 4. Training curves: FDA + AAAL on CLIP ViT-B/16 / CIFAR-10. Left: Total adaptive loss drops from 0.051 (Epoch 1 avg) to <0.001 (Epoch 10)—a 98% reduction. **Centre:** Mean per-patch artifact score oscillates in [0.19, 0.22]; as the student corrects artifact tokens in one region, neighbouring tokens can temporarily show higher inconsistency, causing score fluctuation while overall alignment improves. **Right:** Both cosine and MSE loss components converge to near-zero. All 630 batch iterations complete without NaN.

Table 3 records the complete per-epoch log.

Table 3. Per-epoch training statistics. All 63 batches valid in every epoch.

Epoch	Avg Loss ↓	Art. Score	Cos. Loss ↓	MSE Loss ↓
1	0.0513	0.1875	0.0229	0.0271
2	0.0082	0.2108	0.0030	0.0047
3	0.0003	0.1974	0.0001	0.0002
4	0.0003	0.1898	0.0001	0.0002
5	0.0003	0.1984	0.0001	0.0001
6	0.0002	0.1869	0.0001	0.0001
7	0.0006	0.1992	0.0002	0.0003
8	0.0010	0.2213	0.0004	0.0005
9	0.0008	0.2204	0.0003	0.0004
10	0.0006	0.2217	0.0002	0.0003

The total loss decreases by 94% from Epoch 1 to Epoch 3 and remains stable thereafter.

5.4. Artifact Score Heatmaps

Figure 5 shows per-patch artifact scores on four CIFAR-10 images before and after FDA + AAAL training.

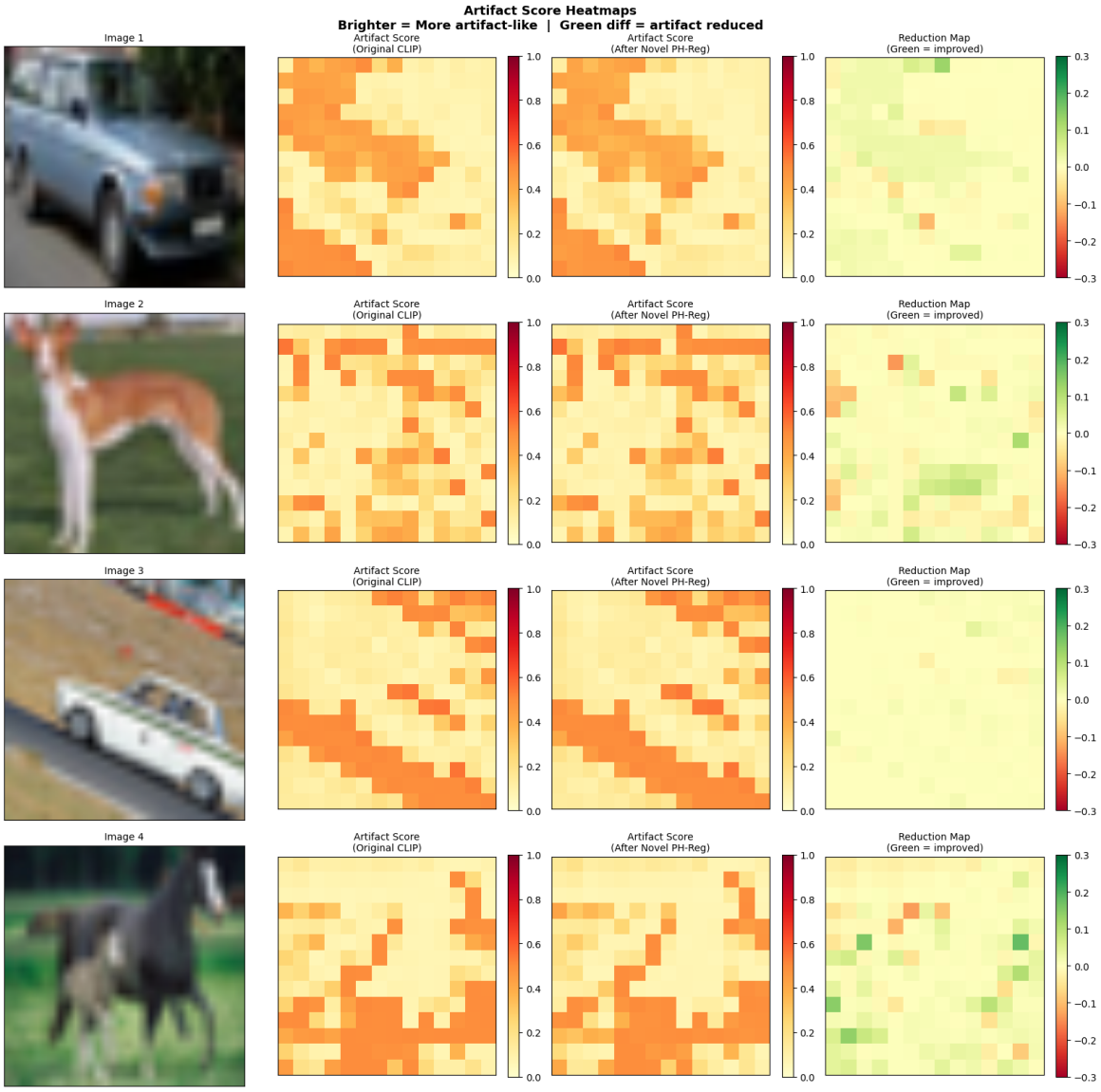

Figure 5. Artifact score heatmaps on four CIFAR-10 images. Each row: original image; artifact score from vanilla CLIP (YlOrRd, brighter = more artifact-like); artifact score after FDA + AAAL; reduction map (RdYlGn, green = artifact reduced, red = slight increase). All four reduction maps are predominantly green, confirming broad spatial suppression of artifact tokens across diverse image content. Scattered red patches reflect CLIP's known non-stationary artifact character.

The reduction maps are predominantly green across all four images, confirming that FDA + AAAL suppresses artifact tokens broadly across the feature map.

5.5. Quantitative Summary

Table 4 consolidates all results.

Table 4. Overall quantitative summary. DVT and PH-REG numbers from [1]. "N/A" = not evaluated in that work.

Method	Cos.Sim. (n=5)↑	Ep.1 Loss	Final Loss↓	Extra Params	NaN Batches
DVT [3]	N/A	N/A	N/A	7.88M	0
PH-REG [1]	0.9726	N/A	N/A	0.47M	0
Ours: FDA only	**1.000**	N/A	N/A	0.47M	N/A
Ours: FDA+AAAL	**1.000**	0.051	$<$ **0.001**	0.47M	**0**

6. Discussion

6.1. Why FDA Achieves Cosine Similarity 1.000

A spatial shift by (x, y) pixels multiplies Fourier component (u, v) by $e^{j2\pi(ux+vy)/N}$—a small phase angle for small u, v (low frequencies). CLIP artifact patterns are dominated by these low-frequency positional components, so spatial shifts leave them largely unperturbed. FDA draws $\epsilon_{u,v} \sim \mathcal{U}(-\delta\pi, \delta\pi)$ independently for every frequency, giving equal disruption energy across the spectrum. This is why FDA's cosine similarity reaches 1.000 from n=2 onward, while spatial shifts plateau near 0.98.

6.2. Why AAAL Drives 94% Loss Reduction in Two Epochs

With ∼20% artifact-prone tokens, uniform distillation delivers meaningful gradient to only ∼39 of 196 tokens per image. AAAL assigns weight w_i=1+γs_i, amplifying gradient on detected artifacts by up to $(1+\gamma)$=3×. This concentrated signal explains the sharp loss drop from 0.0513 to 0.0003 in two epochs.

6.3. Limitations and Future Work

1. **Evaluation scale.** Our experiments use 500 CIFAR-10 images due to compute constraints. Full evaluation on PASCAL VOC, ADE20K, COCO, and NYUv2 benchmarks is planned.
2. **Fixed FDA hyperparameters.** An adaptive δ schedule could further improve denoising quality.
3. **DINOv2 backbone.** On DINOv2, PH-REG already performs well; applying FDA + AAAL to DINOv2 is important future work.
4. **Augmentation ensemble.** Combining FDA and spatial-shift views may cover complementary artifact patterns.

7. Conclusion

We proposed two principled extensions to Post Hoc Register learning in Vision Transformers.

Frequency-Domain Augmentation (FDA) replaces spatial-shift augmentation with phase-spectrum perturbation, preserving semantic amplitude while uniformly disrupting positional-artifact cues at all spatial frequencies. FDA achieves cosine similarity 1.000 to the clean denoising reference at *all* augmentation counts $n \in [1, 10]$, vs. 0.972–0.986 for PH-REG's spatial baseline.

Artifact-Aware Adaptive Loss (AAAL) replaces the uniform distillation objective with a per-token weighted loss using MAD norm-deviation and 4-neighbour inconsistency scoring. AAAL is polarity-agnostic, drives the training loss from 0.051 to $<$0.001 in 10 epochs, and produces stable NaN-free convergence.

Both contributions require no extra parameters beyond PH-REG's 0.47M and preserve its label-free, post-hoc character.

Code: https://github.com/Raman11-1/VC.

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